M. Reza Skandari, Ph.D.

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Academic and Professional Experience

Academic Experience

2018 – *now*. **Imperial College London.** Imperial Business School, Centre for Health Economics and Policy Innovation. *Assistant Professor of Health Analytics & Operations*

2016 – 2018 **University of Chicago.** Department of Medicine, Center for Chronic Disease Research and Policy.
Senior Research Analyst, Faculty Development Program Instructor

Education

2011 – 2016 **University of British Columbia** PhD in Management Science.

2009 – 2011 **University of Florida** M.Sc. in Operations Research.

2007 – 2009 **University of Tehran** M.Sc. in Industrial & Systems Engineering.

2003 – 2007 **Sharif University of Technology** B.Sc. in Industrial & Systems Engineering

Research Interests

Interpretable AI applications in health-care and clinical settings

AI methodologies: large multimodal models including LLMs and RAGs, reinforcement learning

Mathematical methodologies used in optimizing stochastic dynamic systems

Operations efficiency in resource-constrained healthcare systems using data-driven models

Academic and Professional Development

2023 **International Teachers Programme.** Vlerick Business School, Belgium

2020 **Interactive Teaching Workshop.** Educational Development Unit at Imperial.

2019 **Unconscious Bias Workshop.** Imperial.

Research Works

Published Articles

* indicates a student or post-doc I co-authored with

- Morton*, S., Manek, R., Mohammadi, N., **Skandari**, **R.**, & Mullington, C. J. (2026). Quantification of emergency operating theatre process performance: A systematic review of clinical studies. *British Journal of Anaesthesia*. doi:10.1016/j.bja.2025.12.053
- Luo*, Y., **Skandari**, **R.**, Martinez, C., Kilic, A., & Padman, R. (2025). Benchmarking Waitlist Mortality Prediction in Heart Transplantation Through Time-to-Event Modeling Using a New Longitudinal UNOS Dataset. *Proceedings of the AMIA Annual Symposium*. arXiv: 2507.07339 [stat.AP].
  Best student paper finalists

- Winn, A. N., Newman, Z., Deckard, A., Franco-Galicia, M. I., Staab, E. M., Peek, M. E., … Laiteerapong, N. (2025). Development and Internal Validation of the Multiethnic Type 2 Diabetes Outcomes Model for the U.S. (DOMUS). *Diabetes Care*, *48*(11), 1942–1950. doi:10.2337/dc25-0911.
  Impact Factor 16.6, top-ranked journal in diabetes research
- Dadwani*, R. S., Wan, W., **Skandari**, **R.**, & Huang, E. S. (2023). Expected health benefits of sglt-2 inhibitors and glp-1 receptor agonists in older adults. *MDM Policy & Practice*, *8*(2), 23814683231187566. doi:10.1177/23814683231187566
- Porter*, J., Boyd, C., **Skandari**, **R.**, & Laiteerapong, N. (2023). Revisiting the time needed to provide adult primary care. *Journal of General Internal Medicine*, *38*(1), 147–155. doi:10.1007/s11606-022-07707-x
- Nair*, V., Auger, S., Kochanny, S., Howard, F. M., Ginat, D., Pasternak-Wise, O., … Pearson, A. T. (2022). Development and Validation of a Decision Analytical Model for Posttreatment Surveillance for Patients With Oropharyngeal Carcinoma. *JAMA Network Open*, *5*(4), e227240–e227240. doi:10.1001/jamanetworkopen.2022.7240
- Choi*, J. G., Winn*, A. N., **Skandari**, **R.**, Franco, M. I., Staab, E. M., Alexander, J., … Philipson, L. et al. (2022). First-line therapy for type 2 diabetes with sodium–glucose cotransporter-2 inhibitors and glucagon-like peptide-1 receptor agonists: A cost-effectiveness study. *Annals of Internal Medicine*, *175*(10), 1392–1400. doi:10.7326/M21-2941.
  Impact Factor: 15.3, top general medicine journal
- Alexander, J. T., Staab, E. M., Wan, W., Franco, M., Knitter, A., **Skandari**, **R.**, … Philipson, L. H. et al. (2021a). Longer-term benefits and risks of sodium-glucose cotransporter-2 inhibitors in type 2 diabetes: A systematic review and meta-analysis. *Journal of General Internal Medicine*, 1–10. doi:10.1007/s11606-021-07227-0
- Alexander, J. T., Staab, E. M., Wan, W., Franco, M., Knitter, A., **Skandari**, **R.**, … Philipson, L. H. et al. (2021b). The longer-term benefits and harms of glucagon-like peptide-1 receptor agonists: A systematic review and meta-analysis. *Journal of General Internal Medicine*, 1–24. doi:10.1007/s11606-021-07105-9
- Skandari, M. R., & Shechter, S. M. (2021). Patient-type bayes-adaptive treatment plans. *Operations Research*, *69*(2), 574–598. doi:10.1287/opre.2020.2011
- Dadwani*, R., **Skandari**, **R.**, GoodSmith, M., Phillips, L., Rhee, M., & Laiteerapong, N. (2020). Alternative type 2 diabetes screening tests may reduce the number of us adults with undiagnosed diabetes. *Diabetic Medicine*, *37*(11), 1935–1943. doi:10.1111/dme.14330
- Miksanek*, T. J., **Skandari**, **R.**, Ham, S. A., Lee, W. W., Press, V. G., Brown, M. T., & Laiteerapong, N. (2020). The productivity requirements of implementing a medical scribe program. *Annals of Internal Medicine*. doi:10.7326/M20-0428.
  Impact Factor: 15.3, top general medicine journal
- GoodSmith*, M. S., **Skandari**, **R.**, Huang, E. S., & Naylor, R. N. (2019). The impact of biomarker screening and cascade genetic testing on the cost-effectiveness of MODY genetic testing. *Diabetes Care*, *42*(12), 2247–2255. doi:10.2337/dc19-0486.
  Impact Factor 16.6, top-ranked journal in diabetes research
- Wan, W., Nathan, A. G., **Skandari**, **R.**, Zarei, P., Reid, M. W., Raymond, J. K., & Huang, E. S. (2019). Cost-effectiveness of shared telemedicine appointments in young adults with T1D: CoYoT1 trial. *Diabetes Care*, *42*(8), 1589–1592. doi:10.2337/dc19-0363.
  Impact Factor 16.6, top-ranked journal in diabetes research
- Laiteerapong, N., Cooper, J. M., **Skandari**, **R.**, Clarke, P. M., Winn, A. N., Naylor, R. N., & Huang, E. S. (2018). Individualized glycemic control for us adults with type 2 diabetes: A cost-effectiveness

analysis. *Annals of Internal Medicine*, *168*(3), 170–178. doi:10.7326/M17-0537.
Impact Factor: 15.3, top general medicine journal

- Wan, W., **Skandari**, **R.**, Minc, A., Nathan, A. G., Winn, A., Zarei, P., … Huang, E. S. (2018). Cost-effectiveness of continuous glucose monitoring for adults with type 1 diabetes compared with self-monitoring of blood glucose: the DIAMOND randomized trial. *Diabetes Care*, *41*(6), 1227–1234. doi:10.1177/0272989X18803109.
Impact Factor 16.6, top-ranked journal in diabetes research

- Wan, W., **Skandari**, **R.**, Minc, A., Nathan, A. G., Zarei, P., Winn, A. N., … Huang, E. S. (2018). Cost-effectiveness of initiating an insulin pump in T1D adults using continuous glucose monitoring compared with multiple daily insulin injections: the DIAMOND randomized trial. *Medical Decision Making*, *38*(8), 942–953. doi:10.2337/dc19-0363.
Flagship journal of the Society for Medical Decision Making (SMDM)

- Shechter, S. M., Chandler, T., **Skandari**, **R.**, & Zalunardo, N. (2017). Cost-effectiveness analysis of vascular access referral policies in CKD. *American Journal of Kidney Diseases*, *70*(3), 368–376. doi:10.1053/j.ajkd.2017.04.020.
Flagship journal of the National Kidney Foundation

- **Skandari**, **R.**, Shechter, S. M., & Zalunardo, N. (2015). Optimal vascular access choice for patients on hemodialysis. *Manufacturing & Service Operations Management*, *17*(4), 608–619. doi:10.1287/msom.2015.0552

- Shechter, S. M., **Skandari**, **R.**, & Zalunardo, N. (2014). Timing of arteriovenous fistula creation in patients with CKD: a decision analysis. *American Journal of Kidney Diseases*, *63*(1), 95–103. doi:10.1053/j.ajkd.2013.06.021.
Flagship journal of the National Kidney Foundation

- Salmasi, N., Logendran, R., & **Skandari**, **R.** (2011). Makespan minimization of a flowshop sequence-dependent group scheduling problem. *The International Journal of Advanced Manufacturing Technology*, *56*(5-8), 699–710. doi:10.1007/s00170-011-3206-9

- Azadeh, A., **Skandari**, **R.**, & Maleki-Shoja, B. (2010). An integrated ant colony optimization approach to compare strategies of clearing market in electricity markets: Agent-based simulation. *Energy Policy*, *38*(10), 6307–6319. doi:10.1016/j.enpol.2010.06.022

- Salmasi, N., Logendran, R., & **Skandari**, **R.** (2010). Total flow time minimization in a flowshop sequence-dependent group scheduling problem. *Computers & Operations Research*, *37*(1), 199–212. doi:10.1016/j.cor.2009.04.013

Under Review

- Mohammadi*, N., Pearson, A., & **Skandari**, **R.** (2025). *Multi-model constrained optimization for stratified cancer surveillance strategies*. Under review at *Production and Operations Management (POM)*. Download pre-print from SSRN: 5383166

- Mohammadi*, N., **Skandari**, **R.**, & Shah, A. (2025). *Efficient discovery of cost-effective policies: The case of hearing loss screening for cystic fibrosis*. Under review at *Production and Operations Management (POM)*. Download pre-print from SSRN: 4338022

- Morton*, S., Manek, R., Mohammadi, N., **Skandari**, **R.**, & Mullington, C. J. (2025). *Quantification of emergency operating theatre performance: A systematic literature review*. Under review at *British Journal of Anaesthesia*.

- Zamani*, N., Mohammadi, N., & **Skandari**, **R.** (2025). *Stratify, don't personalize: Balancing implementation complexity and outcome disparity*. Under review at *Manufacturing and Service Operations Management (MSOM)*. Download pre-print from SSRN: 5388447

Dissertations

- **Skandari, R.** (2016). *Optimal Treatment Planning under Consideration of Patient Heterogeneity and Preparation Lead-Time* (Doctoral Dissertation, Sauder School of Business, University of British Columbia, Vancouver, BC, Canada). Under the supervision of Prof. Steven Shechter. doi:10.14288/1.0314138
- **Skandari, R.** (2009). *Designing a Decision Support System (DSS) to Manage Time, Cost, and Quality of a Project.* (Master's Thesis, University of Tehran). Under the supervision of Prof. Reza Ghodsi.
- **Skandari, R.** (2007). *Scheduling a Flowshop Manufacturing Cell with Sequence Dependent Family Setups.* (Bachelor's Thesis, Sharif University of Technology). Under the supervision of Prof. Nasser Salmasi.

Presentations and Seminars

Invited Seminars

- Plenary talk at Markov, Semi-Markov Models and Associated Fields workshop, Sorbonne University, Paris, France. (2025).
- Scool Project, Centre Inria de l'Université de Lille, Lille, France. (2025).
- IMAG, Institut Montpelliérain Alexander Grothendieck, University of Montpellier, Montpellier, France. (2023).
- The Center for Chronic Disease Research and Policy, University of Chicago, (2023).
- European Technology Operations Management (TOM) Seminar Series. (2021).
- Sauder School of Business, University of British Columbia. (2019).
- Department of Industrial & Systems Engineering, University of Wisconsin-Madison. (2018).
- Imperial Business School, Imperial. (2018).
- Industrial & Systems Engineering Department, the University of Tennessee. (2018).
- Lindner College of Business, University of Cincinnati. (2018).
- School of Management, the State University of New York at Binghamton. (2018).
- Systems Science & Industrial Engineering Department, the State University of New York at Binghamton. (2018).
- Desautels School of Management, McGill University. (2016).
- Grand Rounds at Division of Nephrology, University of British Columbia. (2016).
- School of Management, University of California, Merced. (2016).
- University of Alberta School of Business. (2016).

Conference Seminars

- 34th European Conference on Operational Researc. (2025), Leeds, UK.
- 35th POMS Annual Conference. (2025), Atlanta, GA, USA.
- INFORMS Annual Meeting. (2025), Atlanta, GA, USA.
- INFORMS Annual Meeting. (2024), Seattle, WA, USA.
- MSOM Conference. (2024), Minneapolis, USA.
- 33rd POMS Annual Conference. (2023), Orlando, FL, USA.
- INFORMS Annual Meeting. (2023), Phoenix, AZ, USA.

- INFORMS HealthCare Conference. (2023), Toronto, ON, Canada.
- MSOM Conference. (2022), Munich, Germany.
- INFORMS Annual Meeting. (2021), Anaheim, CA, USA.
- Applied Probability Society Conference. (2017), Evanston, IL, USA.
- INFORMS Annual Meeting. (2017), Houston, TX, USA.
- Canadian Operations Research Society Annual Conference. (2016), Nashville, TN, USA.
- INFORMS Annual Meeting. (2016), Nashville, TN, USA.
- INFORMS Annual Meeting. (2015), Philadelphia, PA, USA.
- Manufacturing & Service Operations Management Conference. (2015), Toronto, ON, Canada.
- INFORMS Annual Meeting. (2014), San Francisco, CA, USA.
- Canadian Operations Research Society Annual Conference. (2013), Vancouver, BC, Canada.
- INFORMS Annual Meeting. (2012), Phoenix, AZ, USA.
- The 34th Annual Meeting of the Society for Medical Decision Making. (2012), Phoenix, AZ, USA.
- INFORMS Annual Meeting. (2011), Charlotte, NC, USA.
- INFORMS Annual Meeting. (2010), Austin, TX, USA.

Teaching and Lectureship

Post-graduate Course Leader

2025. **Data Structures and Algorithms** MSc in Business Analytics, Financial Technology. *Imperial Business School*

2019 – 2024. **Decision Analytics** Master of Management Programme. *Imperial Business School*

2021–2022 **Project Management** Master of Management Programme. *Imperial Business School*

2019 **Stochastic Dynamic Programming** Doctoral Programme. *Imperial Business School*

2014 – 2015 **Simulation for Decision Making** Master of Business Analytics Programme. *Sauder School of Business, University of British Columbia*

Advanced Simulation for Decision Making Master of Business Analytics Programme. *Sauder School of Business, University of British Columbia*

Faculty Development Program, Course Leader

2017, 2023 **Cost-Effectiveness Analysis and Modeling** *Department of Medicine, University of Chicago*

Undergraduate Course Leader

2018—2021 **Project Management** The Business for Professionals of Engineering and Science (BPES) Programme. *Imperial Business School*

2015 **Logistics and Operations Management** Bachelor of Commerce (BCom) *Sauder School of Business, University of British Columbia*

Teaching Assistant

2011—2016 **University of British Columbia** Managerial Decision Modeling and Analytics, Logistics and Operations Management

2010—2011 **University of Florida** Quality Control

2007—2009 **University of Tehran** Production Planning and Inventory Management

2003—2007 **Sharif University of Technology** Queuing Theory, Discrete Event Simulation, C Programming Language, Management Information Systems

Grants and Awards

Research Grants

2019, 2023 **Breakthrough T1D**, *Co-Principal Investigator*
Potential Medicare Savings of Enhanced Glucose Control Strategies in Patients with Diabetes

2018-2022 **American Cancer Society**, *Co-Principal Investigator*.
Mathematical Prediction Models to Optimize Post-Treatment Surveillance in Head and Neck Cancer.

Bucksbaum Institute for Clinical Excellence, *Co-Principal Investigator*.
Building the Business Case for Medical Scribes

American Diabetes Association, *Co-Investigator*.
Examining the U.S. cost-effectiveness of alternative glycemic treatment algorithms in the era of new medications

National Institute on Minority Health and Health Disparities, *Co-Investigator*. Predicting and Reducing Future Health Disparities for U.S. Adults with Diabetes

Awards

2022 **Teaching Excellence Award in PhD Supervision**. Imperial Business School, Imperial.

2012 **Lusted Award Finalist**. Society for Medical Decision Making.

2012—2015 **Dean Earle D MacPhee Memorial Fellowship**. Awarded by the University of British Columbia.

2011—2015 **Sauder School of Business Graduate Award**. Awarded by the University of British Columbia.

2010—2012 **Alumni Fellowship**. Awarded by the University of Florida.

Mentorship and Student Development

PhD Students at Imperial

2019 – 2025. **Narges Mohammadi**, Business School
Winner of the President's PhD Scholarship in 2020, Imperial College London

2020 – curr. **Niloofar Zamani**, Business School

Tarita Murray-Thomas, School of Public Health

External PhD Students

2023 – curr. **Yingtao Luo**, Carnegie Mellon University
Joint PhD programme in machine learning and public policy

Master of Research Leading to PhD

2019 – 2020. **Narges Mohammadi**, Business School

2020 – 2021. **Niloofar Zamani**, Business School

Pre-doctoral

2019 – curr. **Jaime Cheong**
NIHR Integrated Clinical and Practitioner Academic (ICA)
Programme Pre-doctoral Clinical and Practitioner Academic Fellowship (PCAF)

Mentorship and Student Development (continued)

Laidllaw Scholarship

2025 – curr. **Hanna Seo**, School of Medicine, Imperial

Lana Wang, Mechanical Engineering Department, Imperial

Academic and Administrative Leadership

Recruitment Activities

2018 – curr. **Faculty hiring committee**
Participated in candidate selection, panel interview, and candidate engagement

2024. **Imperial College Research Fellowship**
Interviewed and assessed postdoc candidates

2023-curr. **The Laidlaw Scholars Leadership and Research Programme**
Selected, interviewed, and mentored successful students

2018 – curr. **PhD Programme**
Involved in outreach, selection, interview and assessment, provided mentoring to admitted candidates

2023-curr. **BSc Economics, Finance and Data Science Programme**
Interview and assessed potential candidates, provided academic mentoring to admitted candidates

Administrative Service

2025 **Staff Survey Action Committee**
Contributed on behalf of Analytics, Marketing and Operations department

Programme Review

2024. **MSc Global Health Management**
Participated in rebranding and programme content review

Scholarly Engagement and Editorial Contributions

Inclusive Global Research Engagement

2022–2024. Elected Board Member, International Council

INFORMS Health Applications Society

Journal Peer Review and Editorial Duties

Operations **Management Science**

Operations Research

Manufacturing & Service Operations Management

Production and Operations Management

Naval Research Logistics

Health Care Management Science

IISE Transactions

Economics **Health Economics**

Clinical **Heart, BMJ, Diabetes Care**

Scholarly Engagement and Editorial Contributions (continued)

Grant Peer Review

Non-UK **NSERC**, Natural Sciences and Engineering Research Council of Canada
NIH, The National Institutes of Health, USA
UK **MRC**, Medical Research Council

Academic Events and Outreach

Conference Cluster Organization

2025. **Healthcare Cluster**
INFORMS, International Conference - Singapore,

Academic Seminar Initiatives

2025–2026. **HAS International Seminar Series** for Postdocs and PhD Students
INFORMS Healthcare Applications Society
Led the ideation, launch, and organization of the series

2023. **ETOM Seminar Series**
The European Technology & Operations Management (E-TOM)
Organized and moderated seminars

Best Paper Committees

2025. Best Healthcare Management Paper
Decision Sciences Institute (DSI)
Assessed papers and made recommendations

2022–2024. Healthcare Special Interest Group (SIG)
Manufacturing & Service Operations Management
Reviewed and selected submissions

Session Organization

2015 – curr. **INFORMS**, Annual Meeting
2015 – 2023. **INFORMS**, Healthcare
2025. Production and Operations Management Society (**POMS**)

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